AgentWorkShop: A Modular Task-to-Device Integration Framework for Industrial Agents

***Abstract*—Industrial agents must translate task goals into device operations while respecting changing production context and scene-specific signals and limits. We present AgentWorkShop, an integration framework that connects task and direct-operation entries to shared dispatch, maps semantic controls to device drivers, and retains command receipts separately from subsequent observations. Simulator checks examine protocol and node access, and two entry paths are tested for their retained records. In a separate mock-driver test with simulated approval decisions, every injected stale production context was rejected before a write; removing the context comparison allowed six product or run changes. Four deterministic simulator missions recorded on September 28, 2026, trace control requests and observations for injection molding, film production, annealing, and wastewater treatment. In one trajectory per scene, each reported final primary value fell within its target band. The tested routes retained entry-specific records, and the missions reached their reported target bands in simulation. The missions do not use a language model to select setpoints; they do not establish human-review effectiveness, statistical repeatability, or physical-plant performance, and some secondary terminal measurements were not archived.**

***Index Terms*—Industrial agents, industrial communication, industrial informatics, production systems, semantic data integration.**

I. Introduction

INDUSTRIAL-AGENT deployments translate task-level goals into operations over production signals and equipment. A line-speed adjustment, for example, depends on the active product, recipe, run, applicable limits, and the control node mapped to the line. If that context changes while a proposal is awaiting approval, a formerly admissible request may no longer be appropriate. Device access also needs evidence at two distinct points: what the command interface reported and what the process was later observed to do. Connecting these steps across task and device interfaces is therefore a central industrial integration problem.

Recent work explores end-to-end automation, control-sequence generation, modular factory skills, and robotic-line control [1]–[4]. Other studies address programmable logic controller (PLC) code generation, context-aware control, resource discovery, shop-floor coordination, digital-twin fault handling, and natural-language machine access [5]–[10]. Recent platforms further connect proposals to Asset Administration Shell (AAS) models and signed approval, validate actions against process and instrumentation diagrams (P&IDs), or organize sensor–manufacturing execution system (MES)–enterprise resource planning (ERP) evidence streams [11]–[13]. Together, these efforts provide valuable methods for agent reasoning, production context, and operation constraints.

These studies address complementary aspects of agent operation and motivate a further integration question: how can different task engines and direct calls use a common operation boundary while accounting for scene-specific devices, production context, and route-specific evidence? AgentWorkShop provides such a design. Registered harnesses connect agent engines to task context, while direct representational state transfer (REST) calls reach the same named-operation dispatcher. Scene blueprints map semantic signals to configured drivers and provide targets and guards. The scope of the present experiments does not establish that this design is unique or that all entry paths are equivalent.

Fig. 1 summarizes this control path. Task goals and production context inform an operation proposal. Shared checks, route guards, approval when required, and a pre-dispatch context recheck precede writes through the device write-control (DCW) subsystem. Command receipts and subsequent data-acquisition (DAQ) observations remain distinct records: a readback can verify a setpoint, while a later observation describes the measured process state without, by itself, establishing causation.

AgentWorkShop implements this path as a modular task-to-device framework. Task requests enter a collaboration runtime whose harness adapters connect agent engines to production context and available operations. Direct calls enter the operation dispatcher without creating a task. Scene blueprints and channel profiles define the signals, objectives, guards, tools, and agent bindings for a deployment. Fig. 2 shows these architectural relationships, and Fig. 3 follows one registered task through the execution path.

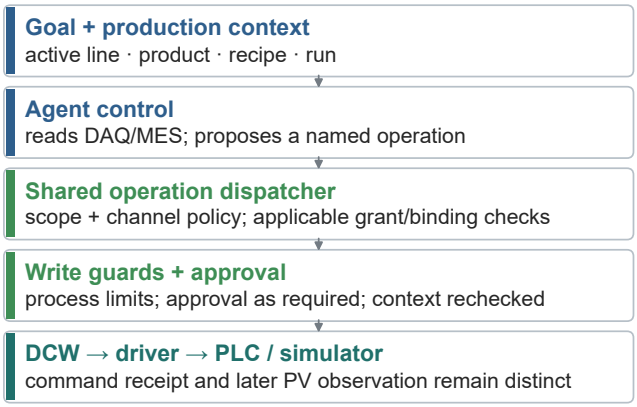


Fig. 1. Agent-control path for a configured industrial task. The agent uses DAQ and MES context, passes shared operation checks, and routes writes through process guards, conditional approval, and a pre-dispatch context recheck. DCW and configured drivers provide device access; command receipts and later process observations remain separate. Scene configuration supplies targets and guards, while the application provides its optimization policy.

This paper contributes a modular task-to-device integration design that combines a shared dispatch boundary, scene-specific signal mappings, and separate command and observation records. It also examines selected parts of that design through complementary tests: simulator protocol and node access, entry-specific records, stale approval-context rejection with a mock driver, and four process missions. In the component test, every single-field context change was refused before mock writes; this result concerns the tested approval check, not human review. The missions trace requests and observations for part weight, film thickness, strip hardness, and dissolved oxygen. Each scene has one deterministic trajectory, and no language model selects its setpoints. The checks do not establish equivalence between entry paths, comparative performance, or physical-plant performance.

Section II reviews related work. Section III presents the architecture and execution flow. Section IV explains the cooperating modules and their device access and evidence mechanisms. Section V describes the simulation protocol and results; Sections VI and VII discuss their implications and conclude.

II. Related Work

This section positions the integration problem against plant-facing agents and industrial abstractions for context, device operations, and operational evidence. Table I summarizes selected studies' reported mechanisms and evaluation settings; it is not a matched performance comparison or an exhaustive account of their implementations.

A. Industrial Agents and Plant Integration

Industrial-agent systems span end-to-end automation control, supervisory services, programmable logic controller (PLC) interfaces, manufacturing coordination, and operator-facing machine access. Xia et al. present a large language model (LLM) framework for end-to-end automation control [1]; Huang et al. focus on automated control generation for industrial cyber-physical systems [2]; Biskupovic et al. study agent-based supervision [14]; and Agents4PLC and InstructMPC address PLC code generation and human–LLM control [5], [6]. Manufacturing studies consider resource-capability exploration [7], shop-floor coordination [8], and digital-twin fault handling [9]. Natural-language machine access [10] and real-time LLM control of a robot-operated line [4] extend this work to operator interfaces and robotic-line control.

Xia et al. connect cognitive, digital-twin, and physical-automation layers and organize twin information as state snapshots, event messages, and plan sequences [15]. Other nearby designs expose factory-module operations through the Model Context Protocol (MCP) with state-based tool filtering, constrain manufacturing plans, or compose semantic recipes from OPC Unified Architecture (OPC UA) skills [3], [16], [17]. Several recent systems also combine agent execution with approval, industrial data, or audit records. Graul et al. ground proposals in the Asset Administration Shell (AAS) and route approved actions through OPC UA Methods, with signed approvals and audit logging [11]. Schall separates advisory language-model agents from process and instrumentation diagram (P&ID)-grounded verification and execution, and evaluates structural checks across five simulated process scenarios [12]; its context includes supervisory control and data acquisition (SCADA) and standard operating procedure (SOP) information. Lin and Chen build a governed sensor–MES–ERP digital thread for closed-loop monitoring and control on real manufacturing records [13]. These designs address different parts of the integration problem: asset-grounded approval, action validation, and the organization of industrial evidence. The present paper describes a complementary task-to-device path in which registered tasks and direct calls reach named operation handlers and route-specific records remain with their owning services. The cited studies' reported descriptions do not permit a matched comparison of that entry-path arrangement; Table I therefore presents their mechanisms and evaluation contexts rather than inferring missing capabilities.

The operation checks complement safety-aware decision support [18] and operate alongside the plant's configured process protection.

B. Production Context, Agent Interfaces, and Automation

Industrial integration also depends on abstractions that are not specific to agent architectures. The Model Context Protocol (MCP) specifies interoperable tool discovery and call mechanisms [19]; the operation-level scope, policy, grant, and binding checks discussed here are AgentWorkShop's own dispatcher rules, distinct from MCP transport authorization. In parallel, ISA-88 and ISA-95 describe recipes and production operations [20], [21]. Run-to-run methods connect process observations to successive production runs [22]. Human–automation research also formalizes the allocation of information acquisition, analysis, decisions, and actions between people and automation [23]. At the agent level, AgentSpec and CaMeL address tool behavior and capability-based information flow [24], [25]. Together, these standards and agent-level mechanisms provide abstractions for production context, tool interaction, and information flow. AgentWorkShop brings these abstractions into one execution path: task and direct-operation calls converge on named handlers, scene mappings connect semantic controls to configured drivers, and owning services retain command receipts and process observations. Section V examines simulated node access, mock approval-context checks, and operation–response sequences in four simulated process scenes. The deterministic missions do not test every entry route or establish the effectiveness of human approval.

III. System Design

AgentWorkShop is designed for industrial tasks whose agents, production contexts, interfaces, and process signals vary between deployments. Its architecture organizes these variations around shared task and operation services, checks each proposed operation against the active production context before device routing, and retains route-specific evidence that distinguishes a command receipt from a measured process response.

TABLE I
REPORTED OPERATION MECHANISMS AND EVALUATIONS IN SELECTED PLANT-FACING STUDIES. EVALUATION SETTINGS AND OUTCOMES ARE SOURCE-SPECIFIC; THIS TABLE DOES NOT COMPARE PERFORMANCE ACROSS STUDIES.

System	Reported operation and evidence structure	Reported evaluation
Xia et al. [15]	Cognitive, twin, and physical layers; twin snapshots, event messages, and plan sequences	Robotic automation and process-simulation cases
Köhle et al. [3]	Injected live state and state-based filtering; MCP-to-OPC UA operations and Message Queuing Telemetry Transport (MQTT) coordination	Nine simulated production challenges
Vieira da Silva et al. [16]	Constrained multi-step plans over MCP manufacturing functions	Laboratory multi-step process
Köhle et al. [17]	Semantic recipes composed from OPC UA skills; virtual commissioning	Unity assembly-workstation proof of concept
Graul et al. [11]	AAS-grounded proposals, signed human approval, PLC-validated OPC UA Methods, and immutable audit log	Two laboratory PoCs; single scripted runs
Schall [12]	P&ID-, SCADA-, and SOP-grounded proposal verification with gated execution	Five simulated scenarios; 43 proposals and 50 LLM runs
Lin and Chen [13]	Governed sensor–MES–ERP digital thread with SOP-, risk-, and capability-aware action gating	25,275 real records; operational metrics
This work	Shared dispatcher and scene-bound DAQ/DCW nodes; separate command and subsequent observation records	Simulator protocol and onboarding checks; mock approval-context test; four deterministic missions

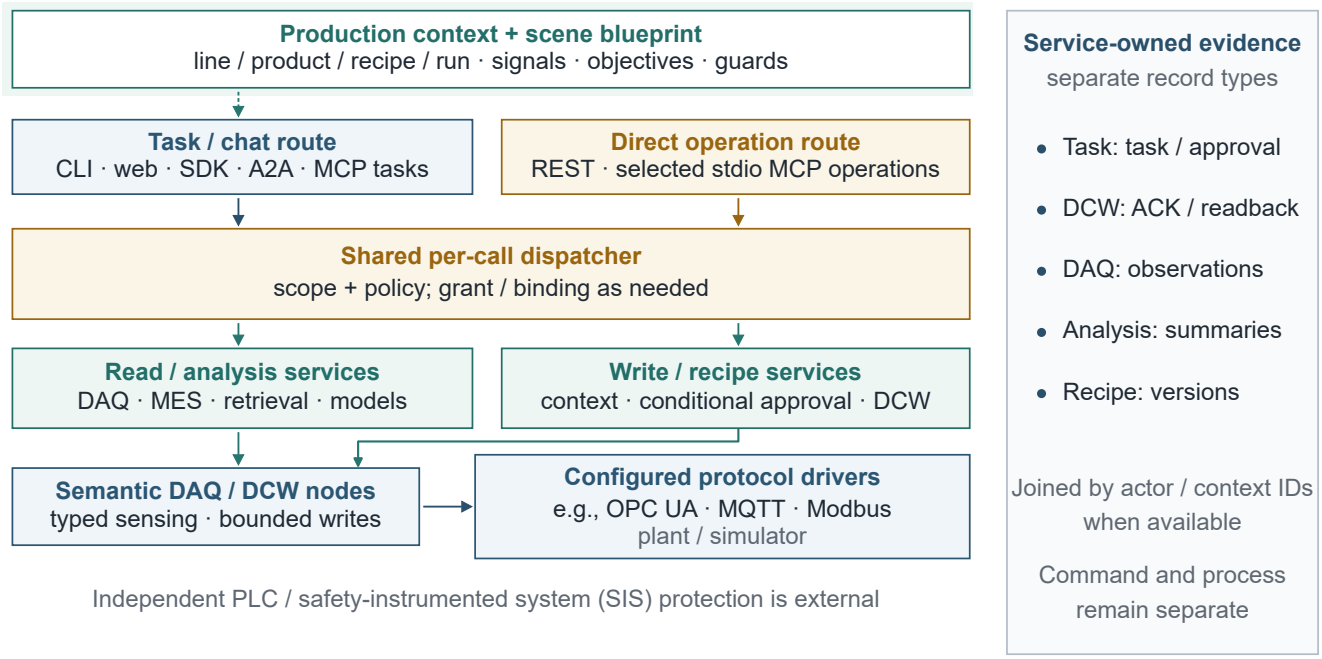


Fig. 2. Architecture and record ownership. Task/chat routes enter the collaboration runtime; direct REST and stdio-MCP calls enter shared dispatch checks without creating a task. One dashed link marks configuration. Write routes apply interval admission and, when configured, approval; recipe dispatch omits the previous-recipe window. The shared dispatcher always checks scope and channel policy, with grant and resource-binding checks applied when required by the operation. Owning services retain distinct records; plant controllers and safety systems retain equipment interlocks.

A. Design Rationale and Requirements

Five requirements—scene variability, harness variability, operation authority, outcome observability, and extensibility—shape the design. Scene variability is represented by blueprints and semantic driver mappings; harness variability by registered adapters; operation authority by shared dispatch checks and route-specific guards and admission; outcome observability by service-owned task, receipt, and observation records; and extensibility by registered interfaces for agent engines and device drivers. Tool visibility is separate from write authorization, and actuator writes converge on the shared DCW admission point.

A scene blueprint bundles the process preset, signal inventory, DAQ/DCW driver mappings, and target and guard rules; judged operation records and later observations remain distinct evidence.

During setup, the integrator binds a validated blueprint to

the active line–product–recipe–run tuple, maps semantic nodes to drivers, and declares target bands and guards. At runtime, the dispatcher checks current authority before the handler issues a command; the handler returns a receipt, and DAQ records subsequent observations.

B. Layered Architecture

Fig. 2 summarizes the access routes, shared dispatch path, production services, semantic I/O, configured drivers, and service ownership of records. Task and chat routes enter the collaboration band, which contains workspaces, channels, task queues, shared memory, mailboxes, task templates, and harness adapters. A *task route* is an execution path through this runtime; a *harness adapter* connects an agent engine to its task execution interface. Direct REST tool calls and the stdio MCP operation bridge bypass task execution in that band (the upper-right direct-operation entry in Fig. 2) and enter the shared governance block without creating a task or starting a harness turn; MCP management and task methods remain on their service routes. Direct calls resolve current route state at invocation, without an agent-turn prompt or a task record. The governance block projects the operation catalog and applies the per-call checks. Write handlers apply admission limits and request approval when the operation and its configuration require it. The context block supplies the active line–product–recipe–run tuple to the gate and scene maps, objectives, and guards to the handlers. The dashed link from that block in Fig. 2 marks this configuration relationship; the runtime assembles prompt-time context from the same records. The service band, behind a route-handler interface that selects an adapter, provides acquisition, write control, analysis, retrieval, and optional model services. Field I/O resolves semantic nodes to built-in or plugin drivers for a plant or simulator.

Each band has a distinct responsibility: task/chat routes enter the runtime, direct operations enter the dispatcher, and both use shared checks. Harness adapters supply agent-specific interaction, semantic nodes hide protocol details, and the driver registry provides the device-facing extension point. Owning services retain task, command, observation, and configuration records separately and link them to actor and production context.

C. Cross-Module Execution Flow

Fig. 3 summarizes a task turn from entry through operation execution, with route-specific evidence retained by owning services. Registered harness adapters receive a scoped tool catalog during setup and supported registry refreshes; the runtime assembles prompt-time production context from current records for each agent turn. For each returned tool call, the shared dispatcher rechecks invocation scope and channel policy and, for industrial operations, current grant and binding conditions before handler selection. The selected handler then applies route-specific guards, active-run checks, approval revalidation, and device-I/O steps as applicable. Command receipts and later process observations remain distinct records.

For an industrial operation u by agent a in channel c, define the shared dispatch condition as

$$\mathcal{A}(a, c, u) = \bigwedge_{X \in \mathcal{C}(u)} X(a, c, u), \tag{1}$$

where S, P, G, and B denote invocation scope, current channel policy, runtime grant, and the agent's binding to referenced resources, respectively. $\mathcal{C}(u) \subseteq \{S, P, G, B\}$ is the set of predicates declared for operation u; it always contains S and P. The grant predicate G is omitted only for an explicitly grant-exempt industrial tool, and B is included when the operation declares resource references. A declared no-reference operation omits B. Each included check must pass before handler selection. The policy predicate includes channel tool selection, profile, and duty-separation checks where configured. Once these predicates pass, the selected route applies its guards and any required approval, then rechecks authority, context, and applicable limits before device dispatch.

For direct REST invocation, the service establishes a and c before applying these checks: an agent token must match the requested agent identifier, or an authenticated user must be the channel owner or an administrator. The manager loads the channel-agent record and rechecks scope and policy; industrial operations then pass the shared grant and binding checks. Handlers resolve live production state from request identifiers and owning services.

Fig. 4 gives reference pseudocode summarizing the registered-harness task path. Direct REST tool invocation bypasses the task runtime and harness but enters the same shared dispatcher. Route-specific timing, approval, and evidence rules are detailed in Section IV.

IV. Module Functions and Integration

The architecture in Section III assembles task context, dispatches operations, and connects device access to separate records of commands and observations. The following subsections explain the handoffs between these functions.

A. Task, Harness, and Context

Task and chat entry points create or join a workspace channel. The task runtime manages membership, task state, and shared execution context; a harness registry adapts each supported agent engine to a common execution interface. At each agent turn, the prompt builder assembles the task, active scene description, production context, channel profile, agent binding, and retrieved memory. This keeps engine-specific interaction behind the harness interface while preserving the current line and operation context for the agent.

Direct REST and selected stdio-MCP operation calls reach the same dispatcher without a task or harness turn. REST authentication and identity resolution follow Section III-C; each handler reads live route state at invocation. Handlers receive the resolved actor, channel, and operation arguments through the same interface for either entry route; records are keyed by the identifiers supplied in each request. A registered task additionally receives prompt context and a scoped tool catalog. Catalog visibility alone does not grant write authority: the dispatcher checks authority at invocation.

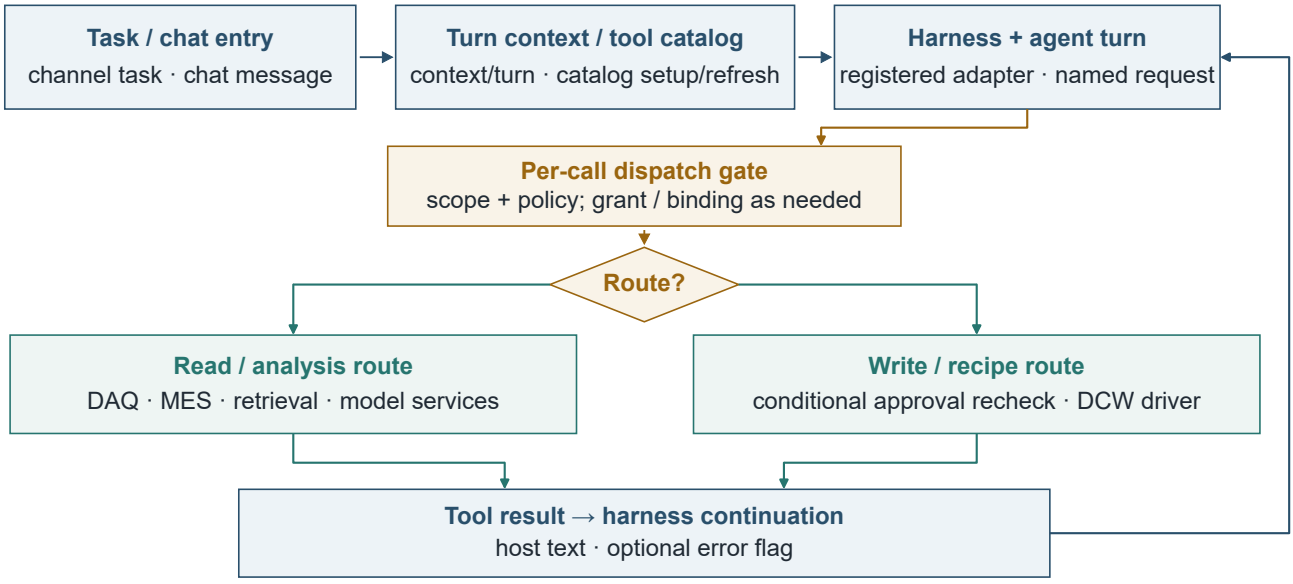


Fig. 3. Registered-harness task flow. The runtime invokes the adapter with a scoped catalog. Scope and channel policy are checked for every call; grant applies to non-exempt industrial operations, and binding applies when resource references are declared (figure label: scope + policy; grant / binding as needed). Failed checks return tool errors. The read path retrieves data. The write path applies route guards, conditional approval and its context revalidation, DCW admission, and receipt grading; a later process observation is separate. Results return to the harness. Device-level interlocks remain with the configured plant-control layer.

Require: task q, channel c, bound agent a, registered scoped catalog T
Ensure: task result; any retained evidence remains with its owning service
$x_t \leftarrow \textsc{ResolvePromptContext}(c, q)$
$m \leftarrow \textsc{RetrieveMemory}(q)$
$p \leftarrow \textsc{AssemblePrompt}(q, x_t, m)$
$h \leftarrow \textsc{InvokeRegisteredHarness}(a, p, T)$
for all tool request u returned by h **do**
  $g \leftarrow \textsc{CheckScopeAndChannelPolicy}(a, c, u)$
  if not g **then**
    $\textsc{EmitToolErrorToHarness}$(h,u); **continue**
  end if
  if u is an industrial operation **then**
    $v \leftarrow \textsc{CheckApplicableGrantBinding}(a, u)$
    if not v **then**
      $\textsc{EmitToolErrorToHarness}$(h,u)
      continue
    end if
  end if
  $k \leftarrow \textsc{SelectHandler}(u)$
  $r \leftarrow \textsc{ExecuteRouteHandler}(k, a, c, u)$ ▷ handler resolves live state and may persist route-specific records
  $\textsc{EmitDeclaredToolResultToHarness}$(h,r)
end for
return task state and tool results

Fig. 4. Reference pseudocode for the registered-harness task path. Direct operation calls use the shared dispatcher from their own entry point. Each route persists its records through the owning service and returns its declared response to the harness.

B. Shared Dispatch and Device Admission

The dispatcher is the shared operation boundary. As specified by (1), it checks invocation scope and channel policy for every call; grant is skipped only for a grant-exempt industrial tool, and binding applies when the operation declares resource references. Applicable checks precede handler selection. Handlers revalidate route-specific state at execution time, using the current policy and grant for each invocation. Scene blueprints define device and signal maps, objectives, and process guards. Capability profiles and tool policies narrow available operations, while bindings constrain accessible nodes, controls, and recipes. Optional DAQ, MES, analysis, retrieval, optimization, and digital-twin services are invoked through registered handlers. The application supplies its optimization strategy while the framework coordinates operation execution.

Actuator writes converge on the shared DCW admission point. For an online node write, the admissible value interval is

$$\mathcal{I}_n^{\text{online}}(t) = \mathcal{H}_n \cap \mathcal{B}_n \cap \mathcal{P}_n(t) \cap \mathcal{R}_n(t), \tag{2}$$

where $\mathcal{H}_n$ is the hard engineering range and $\mathcal{B}_n$, $\mathcal{P}_n$, and $\mathcal{R}_n$ are the baseline, active-product, and active-recipe limits; the subscripted limits here are distinct from the policy and binding predicates of (1). Missing optional limits impose no restriction; an empty intersection rejects the request and identifies the excluding layer. Recipe dispatch validates the replacement recipe against the hard, baseline, and product limits rather than the previous recipe window. Step, interval, and hold-window checks also constrain each route. These software checks bound commands alongside independent programmable-logic-controller and safety-instrumented protection.

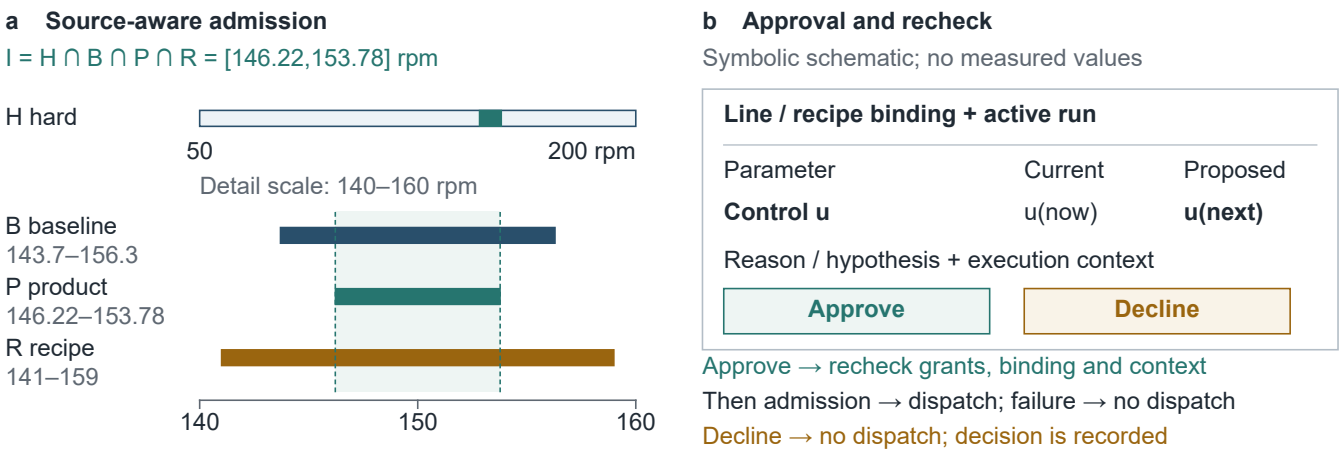


Fig. 5. Admission and approval boundaries. (a) An illustrative rotational-speed configuration in revolutions per minute (rpm): online node interventions use the four-layer interval of Eq. (2), whereas recipe dispatch omits the previous recipe window. (b) Symbolic human approval binds a proposed action and its context tuple; execution rechecks the active line–product–recipe–run context, authority, and applicable limits before device dispatch.

C. Approval, Execution, and Evidence

Routes add timing checks and, when required by operation and configuration, approval checks. Online node writes enforce a 60 s minimum spacing; recipe trials use a per-line cadence (300 s by default), and recipe intervals are anchored to approval. Manual approval binds an actor and proposed operation to candidate values, applicable limits, expiry, and the active line–product–recipe–run tuple. After approval, the handler rechecks authority and compares the current tuple with the approved tuple before entering DCW dispatch. Single-parameter waits expire after 180 s by default; recipe-dispatch approvals use a 30-minute window. A changed recipe definition requires a new `recipe_apply` proposal.

Fig. 5(a) illustrates the four-layer value interval in Eq. (2) using an illustrative rotational-speed configuration in revolutions per minute (rpm). Panel (b) shows a symbolic approval followed by authority and context rechecks; its values illustrate the approval sequence.

A `recipe_trial` illustrates the operation sequence: the handler checks the bound recipe, active run, candidate parameters, and route timing; if required, it requests approval; after revalidation, it submits a bounded batch through the configured driver and returns a per-parameter receipt. The caller obtains later DAQ observations separately and may then update, apply, or roll back a recipe. A trial neither saves nor applies a recipe version. The driver registry maps semantic controls to configured protocol drivers, keeping protocol details out of agent operation arguments.

For a concrete illustration, consider a simulated coating line whose semantic line-speed control is mapped to a Modbus TCP write node. A request to change the setpoint from 37 to 39 m/min is inside a 28–48 m/min recipe window; the remaining applicable limits and timing checks must also pass. If the configured policy requires approval, the proposal binds the candidate and active production context, and the handler rechecks that context before dispatch. The driver then returns a graded receipt. A later acquisition sample is stored separately and can be associated with the operation when the route provides the necessary context and timestamps. This example illustrates the handoffs, not a recorded human decision or a particular process response; the simulator onboarding replay in Section V used test-only automatic approval.

Receipts and observations answer different questions. A command result is graded as readback-verified, transport-acknowledged, or unverified; a readback mismatch can trigger up to three reads at 2 s intervals. Recipe batches retain per-parameter verification outcomes. A readback-verified result records that the returned value matched the request; a transport acknowledgement alone does not establish that the device applied it, and an unverified result does not establish a matching readback. DAQ stores subsequent timestamped process observations, while recipe history, approvals, and task records remain with their owning services. Identifiers and timestamps can link these records when available, but a later observation alone does not prove that the command caused the measured change. Section V evaluates selected parts of this evidence chain under the stated test conditions.

V. Experimental Evaluation

The evaluation examines configured protocol and node access, entry-specific records, approval-context checks, and control requests and observations during optimization. Protocol connectivity and four optimization missions were recorded on September 28, 2026. The missions cover injection molding, biaxially oriented polyethylene terephthalate (BOPET) film production, continuous annealing, and anaerobic/anoxic/oxic wastewater treatment. A separate later onboarding replay and mock-state approval test and two entry-path checks examine separate integration mechanisms; they are not additional trials of the four missions. Table III summarizes the process values, and Fig. 6 shows the recorded intermediate observations.

A. Test Design and Interpretation

The missions ran against locally configured process simulators, using a deterministic scenario-specific control routine and a fixed random seed. The routines inspected process values and requested bounded control changes. The available

benchmark source records an automatic keep judgment after a write returned, before the subsequent process observation; its BOPET routine judges after observing the response. The full executed runtime snapshot was not retained. Annealing and wastewater logs also contain subsequent explicit reversion decisions. These judgments are software events, not evidence of human pre-approval. The records use the direct-node operation interface; they do not exercise all task-entry routes or use a language model to select setpoints. Protocol drivers communicated with simulator endpoints rather than physical equipment. These archived missions illustrate their recorded operation–observation sequence; they do not verify every feature of the architecture described in Sections III and IV.

Each scenario contributes one trajectory. Its initial observation is the within-trajectory reference, not an independent control group. A requested setpoint is an action, whereas a subsequent process value is an observation; the former should not be read as a device readback. The primary outcome is whether the reported final value lies inside the scenario's stated target band. Intermediate observations provide context for the sequence, but are not independent repetitions, and the event positions in the figure indicate order rather than elapsed time.

B. Protocol and Node Integration

The September protocol test configured local simulator endpoints using Modbus TCP, Modbus RTU, OPC UA, MQTT, and HTTP. Its contemporaneous report records a successful connectivity check for each protocol. In a Modbus TCP closed-loop check, seven samples were collected and a 182 °C setpoint was requested; the simulated process reached within 3 °C of that value after 24 s. In a separate fault case, a 185 °C write was read back while the process value remained frozen at 150 °C; monitoring raised an alarm within 2 s. The reported link-fault exercise detected loss of connection and recovered after the simulator restarted. These observations distinguish protocol access and setpoint readback from the later process response. They are from the contemporaneous summary, not a retained complete per-sample protocol trace, and do not establish physical-device behavior.

The BOPET mission used a configured simulated line with nine devices and 49 signals: 30 writable setpoints and 19 measurements. Its four write requests addressed three control nodes. A separate onboarding replay on October 26 instantiated one simulated Modbus TCP device with four write nodes and seven acquisition nodes, together with scalar, vector, table, and image forms from a manufacturing execution system (MES) simulator. The verifier found samples for all seven acquisition nodes, rejected all four out-of-range writes, and confirmed four legal writes by readback; its 23 checks passed. A test-only automatic approval allowed a line-speed change from 37 to 39 m/min, confirmed by the simulator. This single replay demonstrates the configured integration instance, not arbitrary protocol coverage or repeatability. Its source build was not completely pinned, and the MES image used a local storage fallback.

TABLE II
INTEGRATION AND APPROVAL CHECKS. THE PROTOCOL EXERCISE AND ONBOARDING REPLAY USED SIMULATORS; THE APPROVAL TEST USED MOCK DRIVERS AND SIMULATED DECISIONS.

Check	Observed outcome
Protocol access	Five protocol connections succeeded
Node onboarding	All seven acquisition nodes sampled; all invalid writes rejected; all legal writes read back
Approval context	All changed contexts refused before mock writes
Without context comparison	Six product or run changes admitted; all line or recipe changes still refused

C. Task and Direct-Operation Entries

Two separate simulator checks on October 25 exercised a fixed-call registered task and a direct REST operation in each of two scenes. Each issued a recipe trial with synthetic approval and returned a readback-verified receipt and a run-tagged sample. The task route also retained a task–approval–receipt–sample link, although task completion was not asserted; the direct route created no task. Both used the same requested settings, but separate environments. After 20 s, the task and direct samples were both 46.37 μm for film thickness and 674.80 and 675.20 °C, respectively, for annealing temperature; all were below their respective 48–52 μm and 700–760 °C target bands. Each sample was an explicit driver probe stored using a local database fallback with online DAQ collection disabled. These are single probes, not response trajectories, online-pipeline validation, or evidence of attainment. The runs show the records available through each tested entry; they do not establish equivalent behavior or directly trace both calls to the same handler. They are separate from the September optimization missions.

D. Approval Context and Dispatch

An independent component test exercised human-in-the-loop (HITL) approval using simulated decisions and a mock driver. A proposal binds the line, product, recipe, and active run to the approval context; after approval the handler compares that context with the current one before dispatch. The test changed each of the four fields for recipe application, trial, and rollback: all injected stale contexts were refused before mock writes. A further injected change between authorization revalidation and the final check was also refused. An unchanged approved context produced the expected mock writes, whereas a rejected proposal produced none.

To examine the contribution of the context comparison, a test-only ablation removed that comparison while retaining the other checks. Six product or run changes then reached the mock driver and each produced two writes and an execution record. The remaining six line or recipe changes were still blocked by other checks. Thus the comparison supplied the tested product/run protection in this component setup; the result is not a measurement of operator effectiveness, human decision time, physical safety, or behavior during network and device I/O. Table II summarizes the integration and approval checks alongside the separate optimization results in Table III.

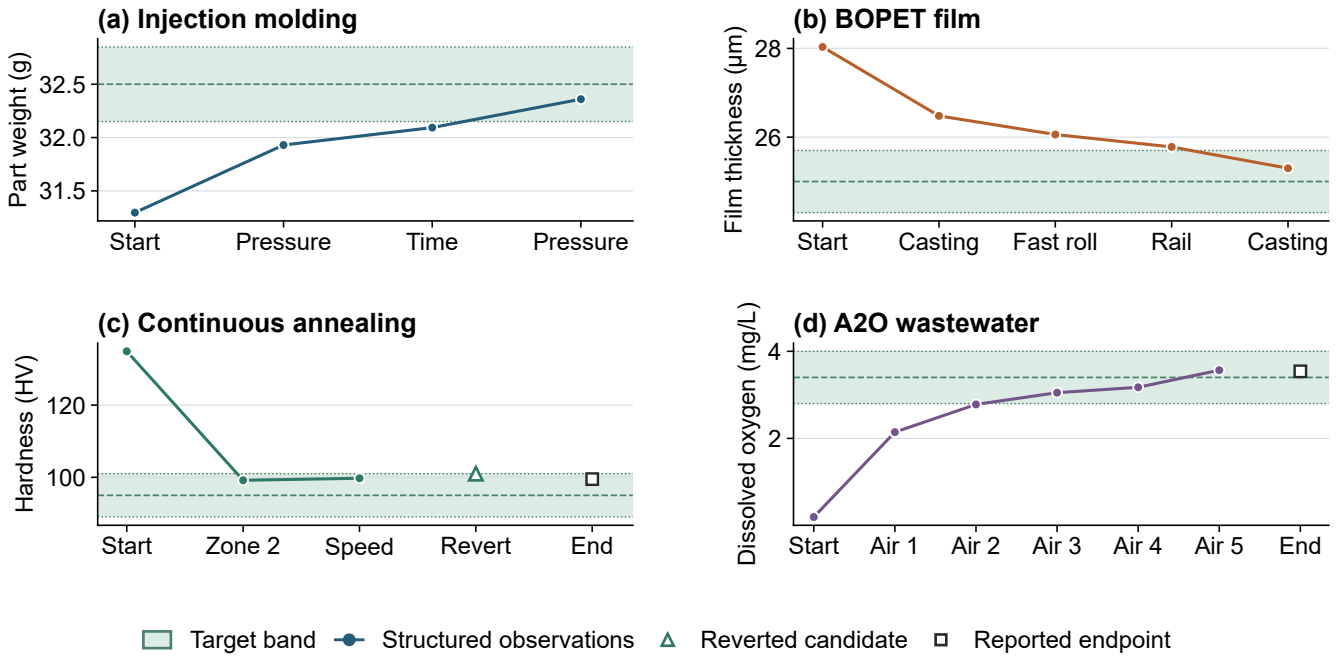


Fig. 6. Process observations from four simulated missions. Shading denotes the scenario-specific target band and circles denote structured trajectory observations. The open triangle marks the annealing candidate that was reverted; its value comes from event text rather than a structured trajectory record. Open squares show separately reported mission-final values for annealing and wastewater; these are not connected to the preceding observations. Horizontal positions indicate event order, not elapsed time. Each panel represents one deterministic trajectory.

TABLE III
PRIMARY PROCESS VALUES IN THE FOUR SIMULATED MISSIONS. EACH ROW IS ONE TRAJECTORY; THE FINAL VALUE IS THE MISSION-REPORTED VALUE.

Scenario	Target band	Initial → final
Injection	Part weight: 32.15–32.85 g	31.296 → 32.36 g
Film	Thickness: 24.3–25.7 μm	28.03 → 25.30 μm
Annealing	Hardness: 89–101 HV	134.9 → 99.5 HV
Wastewater	Dissolved oxygen: 2.8–4.0 mg/L	0.194 → 3.54 mg/L

E. Injection Molding and Film Thickness

The injection-molding mission sought a part weight of 32.5 ± 0.35 g. Starting at 31.296 g, the routine requested an increase in holding pressure from 45 to 57.73 bar, then an increase in holding time from 6 to 10.56 s. The corresponding observations rose to 31.93 and 32.09 g. A further pressure request to 62 bar preceded the final 32.36 g observation, inside the target band. The record therefore shows three linked adjustments in one run; it does not isolate the effect of each parameter through a controlled comparison.

The BOPET mission sought 25.0 ± 0.7 μm film thickness. Its initial observation was 28.03 μm. Four requests changed casting speed, fast-roll speed, rail width, and casting speed again. The successive observations were 26.48, 26.06, 25.78, and 25.30 μm. Thus the final value entered the target band after actions on three controls; the first three observations remained above the 25.7 μm upper limit. The reported melt-temperature observations during the mission were 291.3–291.8 °C, within the configured 268–300 °C window. These measurements characterize this simulated sequence; they do not establish a general tuning rule for film lines.

F. Annealing and Wastewater Treatment

The annealing mission combined a hardness target of 95 ± 6 HV with a line-speed increase. The initial hardness was 134.9 HV. After a zone-2 temperature request from 710 to 850 °C, a structured observation was 99.18 HV. A speed request from 140 to 152 m/min was followed by 99.74 HV. A later 164 m/min candidate was reverted; the event text reports 101.0 HV for that candidate, at the upper boundary of the target band. The mission summary separately reported a final hardness of 99.5 HV and retained speed of 152 m/min. The latter is 8.57% above the initial 140 m/min setting, not a measured production-throughput gain. The rejected candidate reached the upper boundary and failed the configured hardness margin rule, illustrating why the sequence retained the lower request. The record also contains a failed zone-3 read before the zone-2 action. Final tensile-strength and surface-defect measurements were not archived, so the record cannot establish that all annealing quality constraints were met.

The wastewater mission first targeted dissolved oxygen (DO) at 3.4 ± 0.6 mg/L. Five blower-frequency increases from 26 to 34.1 Hz accompanied a rise from 0.194 to 3.567 mg/L in the structured aeration sequence. The routine then adjusted alkali and coagulant dosing for pH and phosphorus, followed by cost-trimming actions. Recorded total-phosphorus observations fell from 0.662 to 0.5507, 0.4607, and 0.413 mg/L during dosing. A later blower reduction was reverted after the recheck found insufficient margin. The mission separately reported final DO of 3.54 mg/L and a change in its simulator-defined cost index from 182.6 to 180.2. This index is not a measured operating cost. The complete final effluent vector was not archived; therefore the reported DO value and cost

index do not independently verify all discharge constraints.

G. Evidence Synthesis

The reported final primary variable lies within its target band in each mission. The recorded paths differ: molding and film production combine several actuators, annealing retains a lower-speed setting after a reverted candidate, and wastewater treatment moves from aeration to chemical dosing. These examples illustrate how requests and observations can be interpreted across different simulated objectives. The protocol exercise, onboarding replay, entry-path checks, and mock approval test address distinct integration and control questions; their counts are not combined with the four process trajectories. Together they do not establish repeatability, causal effects of individual actions, comparative performance, human-approval effectiveness, or transfer to a physical plant.

VI. Discussion

A. Interpretation

The protocol checks and onboarding replay address different steps in device integration. Protocol connectivity demonstrates communication with the tested simulator endpoints, while the replay checks whether one configuration exposes its declared acquisition and control nodes. Neither establishes that an arbitrary device can be integrated without adaptation. The entry-path checks show which records are available for each tested call; separate environments and absent handler-level traces prevent a claim of route equivalence or direct verification of a shared handler. In the separate approval-context test, stale product and run identities reach mock writes when the tuple comparison is removed; their rejection with the comparison enabled therefore isolates the check's contribution within that test setup.

The four missions make the distinction between a control request and a process outcome concrete. In the molding and film examples, several requests precede the final in-band observation; in annealing, the record also contains a reverted candidate. In wastewater treatment, the aeration trajectory and the later mission summary refer to different observations. Keeping these records distinct permits the sequence to be inspected without treating a command as proof of a desired process response.

The missions also illustrate the role of scene-specific configuration. Part weight, film thickness, hardness, and dissolved oxygen have different units, target bands, and available actuators. The operation–response structure accommodates these differences, but the trajectories do not identify which architectural component caused an outcome. Related approaches likewise connect industrial operations to semantic skills, equipment context, or process validation [11], [12], [17]. The present study complements those designs by describing a shared operation path and showing recorded responses in four simulated scenes; the studies use different tasks and do not provide a matched performance comparison.

B. Scope and Limitations

The optimization evidence comprises one deterministic simulated trajectory per scene, without independent replicates or a matched alternative controller. It cannot establish statistical repeatability, superiority to another platform, or the effect of individual controls. The missions use direct-node operations and do not test the full range of entry routes or language-model decisions described by the architecture. Final tensile-strength and surface-defect readings for annealing and the complete final wastewater effluent vector are missing, limiting conclusions about secondary constraints. The process model and simulator configuration also limit transfer to physical production.

The mission records show automated judgments, not human pre-approval. The approval-context test uses simulated decisions and mock drivers; it tests a software condition, not operator performance or the effectiveness of human review [26], [27]. Software operation restrictions are not a complete security or safety assessment [28]–[30]; independent plant protection remains necessary [31]. Future work should repeat missions across initial conditions, preserve all terminal quality measurements, compare matched controllers and entry paths, and evaluate the system with suitable physical equipment and operator oversight.

These results separate three integration decisions: which entry a caller uses, which scene mapping an operation resolves, and which evidence an owning service retains. This separation lets an application change an agent engine or a device mapping without redefining the operation record. The evidence supports this as a software structure; it does not measure engineering effort, operating cost, or reliability gains, which would require matched deployments.

VII. Conclusion

AgentWorkShop connects industrial task context and scene-specific device operations through shared dispatch, configured drivers, and separately retained command and observation records. Simulator connectivity and one onboarding replay exercise configured protocol and node access; separate entry-path checks record route-specific evidence, and a mock test checks stale approval-context rejection. Four local simulation missions illustrate the operation–response sequence across different process objectives; their reported final primary values lie within their respective target bands. The examples show how process evidence can be interpreted alongside requested actions, while the single deterministic trajectories and missing secondary measurements leave repeatability, comparative benefit, human-review effectiveness, and physical-plant performance for future evaluation.

Data Availability: Supplementary File S1 contains the route and integration-check records, the onboarding replay log, the archived mission traces, and hash manifests. The manuscript source, vector figures, and figure-source data accompany the submission package. No physical-plant data are included.

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